The Journal of Nutrition xxx (xxxx) xxx

THE JOURNAL OF NUTRITION

journal homepage: https://jn.nutrition.org/



Nutrition and Disease

Vitamin D Deficiency Is Associated with Increased Uterine Fibroid Risk: A Systematic Review and Meta-analysis

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A B S T R A C T

Uterine fibroids, the most common benign tumors in women of reproductive age, are the leading cause of hysterectomies and disproportionately affect African American women. This is an updated systematic review that performs a novel meta-analysis of observational studies to quantify the association between serum vitamin D concentrations and uterine fibroids across a wide range of studies. In addition, a comprehensive review is included on vitamin D–related factors in both preclinical and clinical studies to assess vitamin D's efficacy in treatment and prevention. Cochrane Library, Embase, PubMed, Scopus, and Web of Science were searched for publications from inception to April 2024. Risk of bias was assessed using Cochrane RoB 2.0 and the Newcastle–Ottawa Scale. Statistical analysis was conducted using R software (R Core Team), and the use of a random-effects model with restricted maximum likelihood estimated the overall effect. Sixty-nine studies met the inclusion criteria and were categorized as preclinical ($n = 26$) or clinical ($n = 43$). One preclinical study explored the downregulation of vitamin D receptors as a mechanism for pathogenesis; others reported a link between vitamin D–related polymorphisms and increased risk of fibroids. Twenty-nine clinical studies evaluated hypovitaminosis D and fibroid prevalence; 26 confirmed an association, whereas 3 found no significant differences. Analysis of 21 clinical studies found a large negative effect size (mean difference = −5.90 ng/mL; $I^2 = 97.4\%$; $P < 0.001$), indicating that women with fibroids had markedly lower vitamin D concentrations compared with controls. Supplementation was shown to improve fibroid-associated symptoms in 9 clinical interventional studies, with some demonstrating a reduction in fibroid size and others showing prevention of fibroid progression. The data support a promising role for vitamin D supplementation in regulating fibroids and highlight the need for further clinical investigation of vitamin D as a potential alternative therapy.

This trial was registered at PROSPERO as CRD420261320154 (20 February, 2026).

Keywords: uterine fibroids, leiomyomas, vitamin D, hypovitaminosis D, vitamin D deficiency, minimally invasive treatment, fibroid treatment, fibroid prevention, fibroid pathogenesis

Introduction

Uterine fibroids are the most common benign gynecological tumor in women of reproductive age [1,2]. This condition presents a major public health issue because it is the leading cause of hysterectomies, with almost 200,000 procedures being conducted annually due to fibroids. In addition, it has been estimated to cost the United States government $44 billion annually when adjusted to 2023 USD [3–5]. Its prevalence varies widely across different epidemiological studies, with rates between 4.5% and 68.6%, depending on the populations studied and the methods used for diagnosis [6]. Moreover, a study from the United States found that most women develop uterine fibroids by age 50, stating an incidence of >80% for

Abbreviations: 25(OH)D, 25-hydroxyvitamin D; 25(OH)D3, 25-hydroxyvitamin D3; ASIP, agouti signaling protein; CI, confidence interval; DHCR7, 7-dehydrocholesterol reductase; EGCG, epigallocatechin gallate; GnRH, gonadotropin-releasing hormone; GWAS, genome-wide association studies; HuLM, immortalized human uterine leiomyoma cells; MD, mean difference; OR, odds ratio; RCT, randomized controlled trial; SNP, single-nucleotide polymorphism; VDR, vitamin D receptor.

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https://doi.org/10.1016/j.tjnut.2026.101838

Received 1 June 2026; Received in revised form 31 August 2026; Accepted 15 September 2026; Available online xxxx

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African American women and nearly 70% for White women [2]. Cases can be slightly symptomatic or asymptomatic, and many times symptoms are downplayed or confused with other conditions. Therefore, the variation in prevalence could be due to the underreporting of the condition, which makes estimations difficult. Still, 25% to 50% of women report symptoms that can range from abnormal uterine bleeding (menorrhagia or hypermenorrhea) to pelvic pain and pressure, and even infertility [1,7]. A study that evaluated the impact of fibroids on quality of life and mental health found that the experience women with fibroids have resembles those of other chronic diseases, stating lower scores than patients with heart disease, diabetes mellitus, and, in some cases, breast cancer in questionnaires assessing vitality, physical function, bodily pain, and social functioning [5].

The exact cause of fibroids is currently unknown; however, they are known to be hormone dependent because they present in women throughout their reproductive years and tend to decline after menopause [2,8]. Given their impact on patients and the healthcare system, understanding the pathophysiology of uterine fibroids is imperative for the development of preventative measures and effective treatments. Hysterectomies and myomectomies are some of the more effective surgical interventions; however, other US Food and Drug Administration–approved options exist, including uterine artery embolization and magnetic resonance imaging–guided focused ultrasound surgery, which are minimally invasive [9–11]. Other minimally invasive treatments have been explored, including collagenase (EN3835), a novel investigational therapy that has been shown to reduce the collagen content in fibroids [12]. Drug treatments currently available for fibroids include leuprolide acetate, a gonadotropin-releasing hormone (GnRH) agonist, and the GnRH antagonists elagolix and relugolix. In addition, other medications, such as ulipristal acetate (a selective progesterone receptor modulator), aromatase inhibitors, and even hormonal contraceptives, have been used to treat symptoms [13–15]. Unfortunately, many of these alternatives involve significant side effects, and their effectiveness at yielding long-term relief of symptoms is still not fully understood. This represents an area of opportunity for developing early detection methods aimed at preventing fibroid incidence in vulnerable populations, as well as for exploring treatments that consider fertility preservation and reducing recurrence rates [14]. Recently, vitamin D has gained attention as a potential low-risk, inexpensive secondary prevention, and therapeutic treatment for uterine fibroids because of its role in various cellular pathways, including the modulation of cell proliferation and the immune system, which may result in inhibitory effects on the growth of fibroids [16].

Vitamin D is a general term that refers to a group of fat-soluble compounds, including ergocalciferol (vitamin D2), found in plants and fungi, and cholecalciferol (vitamin D3), which is produced in higher animals. Vitamin D3 is obtained mainly from sunlight exposure but can also be obtained exogenously through foods like fatty fish [17–19]. Factors, such as geographic location, time spent outdoors, and even skin pigmentation, affect the amount of UV rays reaching the skin and thus the production of vitamin D, which may increase the risk of deficiency in certain demographics [20]. According to the NIH, vitamin D deficiency is defined as $<$ 20 ng/mL 25-hydroxyvitamin D3 (25(OH)D3), whereas the Endocrine Society defines it as $<$ 30 ng/mL [21]. Nevertheless, when defined at $<$ 20 ng/mL 25(OH)D3, deficiency is estimated to affect 41.6% of the United States population and is especially prevalent in historically marginalized groups [22].

Vitamin D is known to play a major role in calcium and phosphate homeostasis, but because of the expression of its receptors in most human tissues, its role also encompasses the regulation of cell growth, differentiation, and apoptosis, ultimately having an antiproliferative effect. Therefore, adequate vitamin D concentrations have been associated with a decreased risk of chronic illnesses, such as different cancers, autoimmune disorders, and cardiovascular disease. As the understanding of the pathology of uterine fibroids continues to expand, studies have described the effect of vitamin D on fibroid growth [23]. In vitro studies have found evidence of inhibiting effects of vitamin D on fibroid growth, whereas rat models treated with vitamin D showed reductions in the size of fibroids [24,25]. In addition, clinical trials have also shown that normalizing vitamin D concentrations halted fibroid growth and, in some cases, decreased the size of existing fibroids [26–28]. There is overwhelming evidence supporting vitamin D as a potential therapy for uterine fibroids, and its low cost, easy accessibility, and limited toxicity makes it an attractive subject for further investment and research efforts. However, there is still a lack of understanding of exactly how vitamin D affects uterine fibroid growth, and because results have differed, there is no consensus on whether it would be useful for prevention, treatment, or both. Therefore, the objective of this systematic review was to examine the association between vitamin D concentrations and fibroid pathogenesis, to assess vitamin D's potential effectiveness in treatment and prevention, with a focused meta-analysis comparing serum 25(OH)D concentrations between women with uterine fibroids and controls. As an extension of the systematic review by Combs et al. [29], this study aimed to gain a larger pool of data to conduct a novel meta-analysis while simultaneously compiling an updated comprehensive overview of the existing evidence surrounding uterine fibroids and vitamin D.

Methods

Eligibility criteria

English-language publications evaluating the relationship between vitamin D and uterine fibroids in humans were considered [30]. Studies using animal models of fibroids were excluded. This is an extension of the systematic review by Combs et al. [29]. In addition to studies previously reported, 22 new studies were included. Studies previously included in the original review were re-evaluated alongside the newly identified studies using the current eligibility criteria. The literature search, study selection, data extraction, and quality assessment were independently repeated for all identified studies.

Information sources

Studies were identified by searching Cochrane Library, Embase, PubMed, Scopus, and Web of Science from inception to March 2022. For the addition of the new studies, the literature search was repeated using the same databases and through manual review from inception to 11 April, 2024. The systematic

review was performed in accordance with the Cochrane Handbook and reported in accordance with the PRISMA guidelines [31]. Recommendations from the Meta-analyses Of Observational Studies in Epidemiology were also followed for the reporting of the meta-analysis results (Supplemental Table 1) [32].

Search strategy

In collaboration with an experienced medical librarian, a detailed list of terms related to uterine fibroids and vitamin D was designed for the literature search using the previously mentioned sources. The search strategies for each bibliographic resource were designed using a combination of database-specific controlled vocabulary and keyword terms, including but not limited to "fibroid," "leiomyoma," and "vitamin D." The complete search strategies used in each database are found in Supplemental Text 1.

Study selection

After initial search strategies were performed, titles and abstracts were screened per the eligibility criteria in 2 rounds by 2 independent reviewers (AGC amd HK) to ensure completeness. Once reviewed, the full-text articles were obtained for studies not excluded to confirm that the complete inclusion criteria were met and full data analysis was available. Any discrepancies were resolved by the senior author (BS). The study included articles that, upon review of the title or abstract, were related to vitamin D and uterine fibroids in humans. Randomized controlled trials (RCTs) and observational studies were included. Studies comprising women with uterine fibroids and fibroid-free controls were included. Studies that evaluated fibroids and vitamin D in animal models without human tissue were excluded. Additional exclusions included studies without an English-language full text, literature reviews, gray literature, dissertations, conference proceedings, scientific abstracts, trial registries, and clinical trial protocols. Studies were then categorized as a preclinical study or a clinical study. The complete eligibility criteria are found in Supplemental Text 2.

Assessment of risk of bias

Study quality was assessed independently by 2 reviewers, and any discrepancies were resolved by the senior author. Observational studies were determined to be good, fair, or poor quality based on the Newcastle–Ottawa scale [30]. Quality was determined by the assessment of selection, comparability, and outcome/exposure categories. RCTs were determined to be either low risk of bias, some concerns, or high risk of bias based on the Cochrane RoB 2.0 tool [33]. Risk of bias was assessed in the domains of the randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result according to guidelines.

Data synthesis

Statistical analysis for the meta-analysis was conducted using version 7.0-0 of meta and version 4.6-0 of the metafor package in R software, version 4.4.0 (R Core Team) [34,35]. Effect sizes were estimated using mean difference (MD) with Hedges' g from studies that compared serum vitamin D concentrations between groups of women with uterine fibroids and control groups of women without fibroids. A random-effects model with a restricted maximum likelihood was used to pool g estimates, and forest plots were used for visualization of these data. Test for heterogeneity was calculated with Cochrane's Q, and statistical heterogeneity among the studies was assessed using I^2 and τ^2. Publication bias and small study effect were assessed using Egger's test, Begg's test, and funnel plots. When there was a significant bias detected in Egger's test and Begg's test, the trim-and-fill method was used to estimate potential missing studies. Sensitivity analyses were performed using the leave-one-out method and influence analysis to assess the robustness of findings, identify potential outliers, and calculate a more precise estimate of effect size. To explore whether the relationship between the variables differed across different subgroups, metaregressions for publication year of study and subgroup analyses by continent were conducted.

Results

Study characteristics

The literature search from the previous study yielded a total of 23 preclinical studies and 24 clinical studies after all the necessary exclusions [29]. The new literature search identified a total of 381 new studies. After screening as per the eligibility criteria and the removal of duplicates, 22 studies fully met the inclusion criteria and were divided into preclinical studies ($n = 3$) and clinical studies ($n = 19$). Together with the original search, there were 26 preclinical studies and 43 clinical studies, for a total of 69 studies (Figure 1). A complete list of the 69 studies included in this paper can be found in Supplemental Table 2, in addition to the study quality of each. Of these studies, 46 were of good quality, 10 of fair quality, 5 of poor quality, 3 were low risk, 2 of some concern, and 3 were high risk (Supplemental Table 2). Next, the results of the included studies are presented in 4 separate sections identified from A to D. Section A includes findings from preclinical studies, section B presents the clinical observational studies, section C reports the data from the meta-analysis, and finally, section D relays the results from clinical interventional studies.

A1. Preclinical studies on human uterine fibroids and vitamin D

Through different in vitro models, the previous review explored potential biological mechanisms by which vitamin D may induce its effect on fibroid growth and development [29]. Seventeen preclinical studies that evaluated the effect of vitamin D on human fibroid cells were discussed, and 1 additional paper was identified from the recent literature search [16,24,36–51]. The studies either used immortalized human fibroid cell lines (primarily HuLM) or uterine primary cells (typically from surgical specimens) or both. Among the 18 included studies, 8 used leiomyoma cell lines and 12 used uterine primary cell lines, with some studies using both types of cells. The added study by Markowska et al. [51] included 110 Polish women, 98 with fibroids and 12 without, and aimed to evaluate the expression of vitamin D receptors (VDRs) through immunohistochemical techniques in 3 different tissue groups: uterine fibroid tissues,

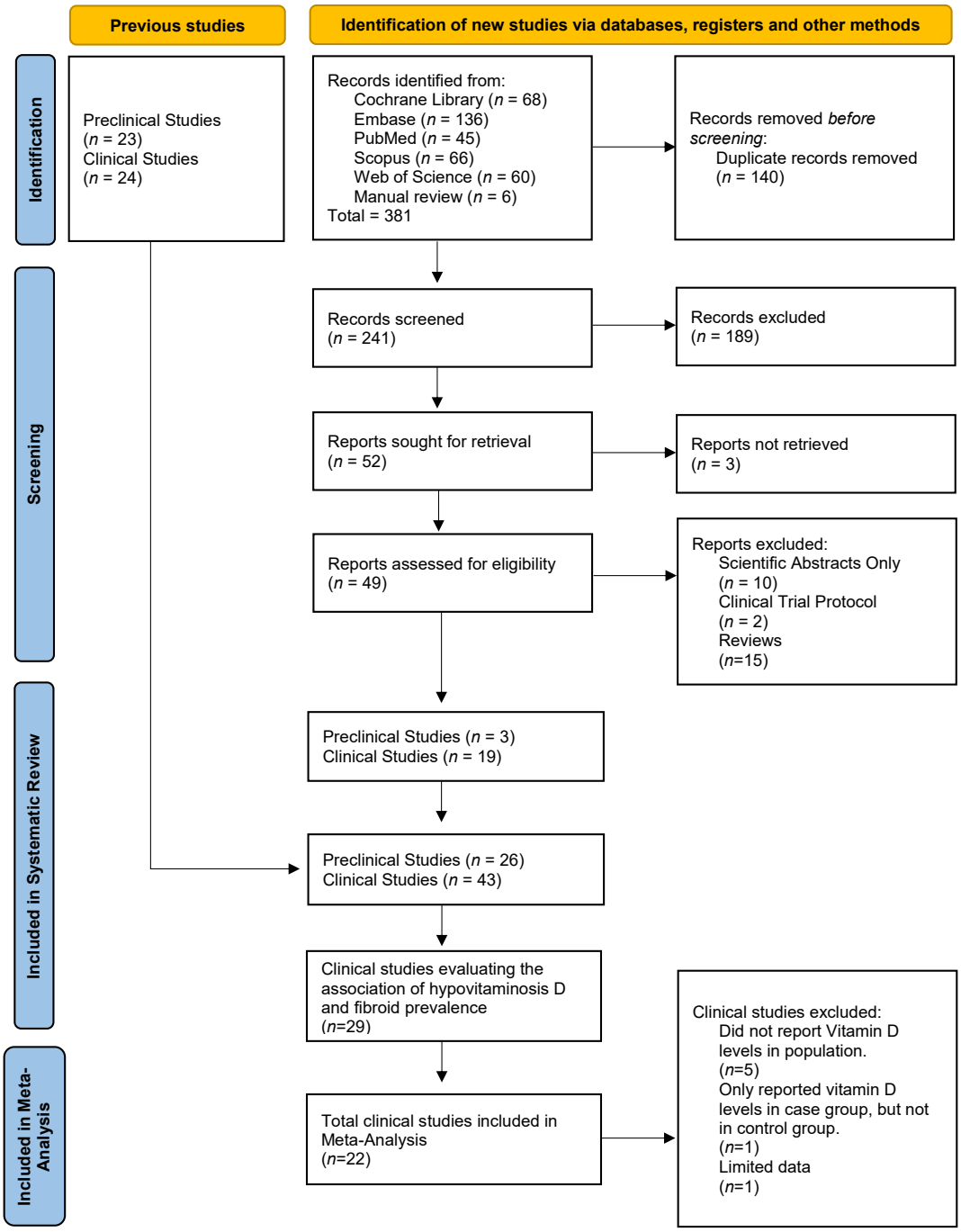


FIGURE 1. PRISMA 2020 study selection flowchart.

uterine tissues from the tumor margin, and normal uterine tissue without fibroids. The findings revealed that the number of immunopositive cells was significantly greater in the peripheries of the fibroids when compared with the fibroids themselves [MD = 190; 95% confidence interval (CI): 80, 300; $P = 0.004$]. However, the number of immunopositive cells in both the fibroids and the peripheries was significantly lower when compared with the normal uterine tissue of the control group (MD = −624; 95% CI: −861, −386; $P < 0.001$ for fibroid) (MD = −434; 95% CI: −672, −200; $P = 0.004$, for periphery). This indicates a significantly lower expression of VDR in fibroids and their peripheries when compared with healthy uterine muscle [51]. This observation supports past findings that link vitamin D to the growth of fibroids and may indicate that the fibroid periphery is an area with a higher potential of developing new fibroids. The other studies explored the impact vitamin D had on cell proliferation, fibrosis, apoptosis, and DNA repair mechanisms through its interaction with various proteins and enzymes [16,24,36–50]. A list of some of the main findings from these preclinical studies can be found in Table 1.

A2. Preclinical studies on genetic polymorphisms of the VDR and uterine fibroids

The previous systematic review mentioned that genome-wide association studies (GWASs) have been able to identify

TABLE 1
Summary of preclinical studies on human uterine fibroids and vitamin D

Topic	Subtopic	Author	Summary
Proliferation	PCNA	Sharan et al. [24]	1 μM vitamin D decreased PCNA protein expression by ~40%
Proliferation	PCNA	Corachán et al. [50]	Vitamin D significantly decreased PCNA expression in human primary uterine fibroid cells.
Proliferation	PCNA	Corachán et al. [46]	Vitamin D at 1 μg/kg/d for 60 d did not significantly reduce PCNA in UF cells in xenograft
Proliferation	Ki67	Corachán et al. [46]	Vitamin D at 1 μg/kg/d for 60 d significantly reduced Ki67 in UF cells in xenograft
Proliferation	Cdk1	Sharan et al. [24]	1 μM vitamin D decreased Cdk1 protein expression 50%
Proliferation	Cdk2	Halder et al. [36]	TGF-β3 (5 ng/mL) induced Cdk2 expression inhibited by vitamin D
Proliferation	Cdk6	Halder et al. [36]	TGF-β3 (5 ng/mL) induced Cdk6 expression inhibited by vitamin D
Proliferation	Cyclin D1	Halder et al. [36]	TGF-β3 (5 ng/mL) induced cyclin D1 decreased by vitamin D3 in a dose-dependent manner
Proliferation	ERK	Sharan et al. [24]	ERK phosphorylation decreased with vitamin D; total ERK expression was unchanged by vitamin D
Proliferation	ERK	Halder et al. [36]	TGF-β3 (5 ng/mL) induced phosphorylation of ERK, which was decreased by vitamin D in a dose-dependent manner Total concentration of ERK was not induced by TGF-β3 or decreased with the addition of vitamin D3
Proliferation	COMT	Sharan et al. [24]	Reduced mRNA and protein expression at 100 nM
Proliferation	ER, PR-A, PR-B	Al-Hendy et al. [39]	Inverse correlation between ER-α, PR-A, PR-B expression, and VDR expression; vitamin D decreased ER-α, PR-A, and PR-B expression
Proliferation	Wnt/β-catenin	Al-Hendy et al. [40]	Vitamin D inhibits β-catenin protein expression in HuLM cells
Proliferation	Wnt/β-catenin	Corachán et al. [45]	Wnt/β-catenin was downregulated in vitamin D–treated human primary uterine fibroid cells based on Wnt gene activity array
Proliferation	Wnt4, Wisp1, FEN1	Al-Hendy et al. [40]	Vitamin D reduced Wisp1 and FEN1 concentrations in HuLM cells; vitamin D decreased Wnt4 nuclear signaling in HuLM and primary uterine fibroid cells
Proliferation	DDIT4, TSC1, and TSC2	Al-Hendy et al. [40]	Vitamin D–induced nuclear DDIT4 in primary uterine fibroid cells; vitamin D increased concentrations of DDIT4, TSC1, and TSC2 in HuLM cells
Proliferation	mTOR and p70S6	Al-Hendy et al. [40]	Vitamin D reduced expression of mTOR in HuLM and primary uterine fibroid cells; Vitamin D reduced concentrations of phospho-p70S6 kinase, but total p70S6 kinase concentrations were unchanged
Fibrosis	TGF-β3	Halder et al. [36]	TGF-β3 (5 ng/mL) induced fibronectin, collagen type-1, PAI-1 expression decreased by vitamin D in a dose-dependent manner
Fibrosis	TGF-β3	Halder et al. [36]	TGF-β3 (5 ng/mL) induced activation of Smad signaling by phosphorylation of Smad2 and induction of complex formation between Smad2, Smad3, and Smad4 inhibited by vitamin D in a dose-dependent manner TGF-β3 (5 ng/mL) induced Smad2 phosphorylation in the nucleus and nuclear translocation of Smad2 and Smad3 reduced by vitamin D
Fibrosis	TGF-β3	Corachán et al. [46]	Vitamin D at 1 μg/kg/d for 60 d reduced TGF-β3 in xenograft
Fibrosis	Collagen type 1	Halder et al. [37]	Vitamin D reduced collagen type 1 protein levels in a concentration-dependent manner in HuLM cells
Fibrosis	Collagen type 1	Corachán et al. [46]	Vitamin D at 1 μg/kg/d for 60 d reduced collagen type 1 in xenograft
Fibrosis	Fibronectin	Halder et al. [37]	Vitamin D reduced excessive fibronectin staining in a concentration-dependent manner in HuLM cells
Fibrosis	Fibronectin	Corachán et al. [46]	Vitamin D at 1 μg/kg/d for 60 d did not significantly reduce fibronectin in xenograft
Fibrosis	PAI-1	Halder et al. [37]	Vitamin D reduced PAI-1 protein expression in a concentration-dependent manner
Fibrosis	PAI-1	Corachán et al. [46]	Vitamin D at 1 μg/kg/d for 60 d reduced PAI-1 in xenograft
Fibrosis	Proteoglycan	Halder et al. [37]	Vitamin D reduced mRNA and protein expression of fibromodulin, biglycan, and versican
Fibrosis	MMP-2 and MMP-9	Halder et al. [38]	Vitamin D reduced MMP-2 and -9 mRNA and protein levels in a dose-dependent manner in HuLM and cultured primary uterine fibroid cells
Fibrosis	TIMP-2	Halder et al. [38]	Vitamin D–induced TIMP-2 mRNA levels and protein levels in a dose-dependent manner in HuLM and cultured primary uterine fibroid cells
Fibrosis	A-kinase anchoring protein 13 (AKAP13)	Cross et al. [48]	Vitamin D reduces expression of AKAP13 mRNA; AKAP13 inhibited VDR activation
Apoptosis	BCL-2	Sharan et al. [24]	Expression significantly downregulated
Apoptosis	BCL-xL	Sharan et al. [24]	No significant change in expression
Apoptosis	BCL-w	Sharan et al. [24]	Expression significantly downregulated
Apoptosis	Apoptosis-Array	Corachán et al. [45]	No significant change in apoptosis in vitamin D–treated cells
Apoptosis	TUNEL assay	Corachán et al. [45]	No increase in percentage of apoptotic cells in the vitamin D–treated group compared with the control group
Apoptosis	Procaspase-3	Corachán et al. [46]	Significantly decreased in the xenograft model

(*continued on next page*)

TABLE 1 (*continued*)

Topic	Subtopic	Author	Summary
Apoptosis	Caspase-3	Corachán et al. [46]	Increased in a dose-dependent manner in the xenograft model
Apoptosis	COX-2	Liu et al. [42]	No significant correlation between VDR and COX—2 expression
DNA repair mechanisms	γ-H2AX	Ali et al. [43]	γ-H2AX was higher in primary uterine fibroid tissue when compared with adjacent myometrium, indicating DNA damage accumulation
Vitamin D concentration	VDR	Markowska et al.[1] [51]	VDR expression was significantly lower in the uterine muscle at the periphery of the fibroid compared with healthy fibroid-free uterine muscle tissue

Abbreviations: AKAP13, A-kinase anchoring protein 13; BCL, B-cell lymphoma; CDK, cyclin-dependent kinase; COMT, catechol-O-methyltransferase; COX-2, cyclooxygenase-2; DDIT4, DNA damage-inducible transcript 4; ERK, extracellular signal-regulated kinase; ER-α, estrogen receptor alpha; FEN1, flap endonuclease 1; MMP, matrix metalloproteinase; mTOR, mechanistic target of rapamycin; PAI-1, plasminogen activator inhibitor-1; PCNA, proliferating cell nuclear antigen; PR-A/PR-B, progesterone receptor A and B; TGF-β3, transforming growth factor-beta 3; TIMP-2, tissue inhibitor of metalloproteinases 2; TUNEL, terminal deoxynucleotidyl transferase dUTP nick end labeling; UF, uterine fibroid; WISP1, WNT1 inducible signaling pathway protein 1; γ-H2AX, phosphorylated histone H2AX.

[1] Newly added studies.

polymorphisms related to vitamin D transport (GC), cholesterol synthesis (DHCR7; 7-dehydrocholesterol reductase), and hydroxylation (CYP241 and CYP24A1), all of which have been associated with affecting the concentrations of circulating vitamin D [29]. Wise et al. [52] explored these genes and others associated with skin pigmentation (i.e., ASIP; agouti signaling protein) and vitamin D and found that Black American women with the GG genotype of single-nucleotide polymorphisms (SNPs) rs12800438 near DHCR7 and rs6058017 in ASIP had a positive association with uterine fibroid risk. Polymorphisms of the VDR were also investigated because of previous GWAS on VDR's influence on cell proliferation, differentiation, and apoptosis. The *Fok1* (rs2228570) polymorphism is a variable C/T transition that is known to alter the physical structure of the VDR protein [53]. Two studies evaluating this polymorphism found that the TT genotype of *Fok1* may correlate with increased fibroid risk [54], whereas the CC genotype may be associated with a reduced risk [55]. In contrast to these studies, Ciebiera et al. [56] found no differences between the groups of women with fibroids and the control with respect to *Taql*, *Bsml*, and *Fokl* polymorphisms. Two new studies examining the effect of vitamin D–related polymorphisms on fibroid risk were identified through this systematic review [57,58]. Xie et al. [57] recruited 900 Chinese women (450 with fibroids and 450 controls) and aimed to examine genetic variants involved in vitamin D metabolic pathways. It found that the vitamin D anabolism–related gene DHCR7 rs1044482 C > T polymorphism was a risk factor for fibroids, especially in patients aged >40 y with a BMI between 18.5 and 25.0. It also examined the NADSYN1 rs2276360 G > C polymorphism, but its relationship with fibroid risk remained unclear [57]. In addition, a Mendelian Randomization study by Guo et al. [58] used a GWAS summary dataset consisting of 441,291 individuals to evaluate the association between vitamin D and the occurrence of uterine fibroids. Ninety-nine SNPs associated with 25(OH)D concentrations were examined, and the study found some of the genetic variants to be associated with a risk of fibroids [58].

A3. Preclinical studies evaluating vitamin D–binding protein and fibroids

The previous review identified 2 preclinical studies that evaluated variability in vitamin D–binding protein (VDBP) and fibroids [59,60]. No new studies were found regarding VDBP and fibroids.

B1. Clinical observational studies evaluating the association of hypovitaminosis D and fibroid prevalence

The previous systematic review by Combs et al. [29] reported findings from 15 clinical studies that evaluated the association of hypovitaminosis D and fibroid prevalence. Of these 15 studies, 14 found significantly decreased vitamin D concentrations in women with fibroids when compared with fibroid-free controls, with only 1 study reporting no association [61–75]. Through the revised literature search, 14 new clinical studies were identified and included in the review. A full list of the 29 clinical studies that reported this association can be found in Table 2. A total of 10,693 participants were included, of whom 3364 had uterine fibroids and 5719 served as fibroid-free control subjects. The remaining 1610 included women who formed part of a prospective cohort, and by the end of the study, 32% had the presence of ≥1 fibroids. In addition, these 29 studies were considered for inclusion in the meta-analysis. Exclusions were made because of reasons described in Supplemental Text 2B and resulted in the inclusion of 22 studies for the meta-analysis (n = 3636 participants, n = 1752 with fibroids, n = 1884 controls). Of the 14 new studies added, 12 yielded results consistent with those previously reported [76–89]. All these studies also used ultrasonography for ascertaining fibroid status, with the exception of Shanmugapriya et al. [78], who did not report the methods used for fibroid characterization or confirmation. Masoem et al. [77] found an association between fibroids and vitamin D deficiency as reported by the significantly lower concentrations of vitamin D in Indonesian women with fibroids when compared with the control group. Four different studies from India also revealed significantly lower concentrations of vitamin D in women with fibroids, and this association was also seen in Nigerian, Egyptian, and Pakistani women [78–81,84,86,89]. Al-azzawi et al. [82] included 70 Iraqi women with fibroids and found that 56% of the participants were deficient in vitamin D (< 20 ng/mL), 25% were insufficient (20–30 ng/mL), and only 19% of the women were within normal concentrations. In addition, Chen et al. [85] measured vitamin D concentrations in 110 Chinese women (55 with fibroids and 55 controls) and found that vitamin D deficiency was more frequent in the group of women with fibroids (50.91% compared with 23.64%); these women also had significantly lower concentrations of vitamin D when compared with the controls. The prospective cohort study by Harmon et al. [83] conducted standardized ultrasounds and measured vitamin D

TABLE 2
List of clinical studies evaluating the association of hypovitaminosis D and fibroid prevalence

Author	Year	Patient cohort	Nationality	Mean D3 (UF vs. control), ng/mL	Odds ratio (OR)
Sabry et al. [61]	2013	$n = 154$ (104 w/UF, 50 control)	Black Egyptian= 87 White Egyptian= 67	Mean D3 (all): 19.7 vs. 22.3 Mean D3 (Black Egyptian) 12.9 vs. 18.3 Mean D3 (White Egyptian) 24.45 vs. 29.53	—
Paffoni et al. [62]	2013	$n = 384$ (128 w/UF, 256 control)	Black Italian= 8 White Italian= 360 Other = 16	Mean D3 (all): 18.0 vs. 20.8	aOR[1] = 2.4
Baird et al. [63]	2013	$n = 1036$ (674 w/UF, 362 control)	Black American = 620 White American = 416	—	aOR[2] = 0.68
Mitro and Zota [75]	2015	$n = 3590$ (469 w/UF, 3121 control)	Black American= 808 White American = 1896 Mexican American = 886	—	White women only: aOR[3] = 2.17
Ciebiera et al. [64]	2016	$n = 188$ (105 w/UF, 83 control)	Polish	Mean D3: 21.9 vs. 26.7	—
Ingala et al. [9] [76]	2016	$n = 432$ (216 w/UF, 216 control)	Congolese	Mean D3: 9.00 vs. 9.10	—
Masoem et al.[9] [77]	2017	$n = 42$ (21 w/UF, 21 control)	Indonesian	Mean D3: 6.70 vs. 10.34	—
Oskovi et al. [65]	2018	$n = 124$ (68 w/UF, 56 control)	Turkish	Mean D3: 6.54 vs. 8.18	—
Singh et al. [66]	2019	$n = 144$ (72 w/UF, 72 control)	Indian	Mean D3: 10.8 vs. 22.9	OR[4] 4.64
Srivastava et al. [67]	2020	$n = 90$ (45 w/UF, 45 control)	Indian	Mean D3: 15.1 vs. 26.1	—
Farzaneh et al. [68]	2020	$n = 148$ (71 w/UF 77 control)	Iranian	Mean D3: 21.4 vs. 24.62	—
Li et al. [69]	2020	$n = 546$ (279 w/UF, 267 control)	Chinese	Mean D3: 18.2 vs. 20.7	—
Irnazarova et al. [70]	2020	$n = 152$ (102 w/UF, 50 control)	Turkish	Mean D3: 16.7 vs. 30.2	—
Sarbhai et al. [71]	2021	$n = 150$ (75 w/UF, 75 control)	Indian	Mean D3: 12.6 vs. 19.0	—
Tunau et al. [72]	2021	$n = 100$ (50 w/UF, 50 control)	Nigerian	Mean D3: 10.2 vs. 14.4	—
Huseman-Plascencia et al. [73]	2021	$n = 106$ (50 w/UF, 56 control)	Mexican	—	OR[5] 2.26
Xu et al. [74]	2021	$n = 213$ (133 w/UF, 80 control)	Han Chinese	Mean D3: 12.1 vs. 16.8	OR[6] 0.8
Shanmugapriya et al.[9] [78]	2021	$n = 100$ (60 w/UF, 40 control)	Indian	Mean D3: 23.6 vs. 45.4	—
Makwe et al.[9] [79]	2021	$n = 88$ (44 w/UF, 44 control)	Nigerian	Mean D3: 34.23 vs. 37.06	—
Alansary and Abd El Megeed[9] [80]	2021	$n = 150$ (75 w/UF, 75 control)	Egyptian	Mean D3: 13.39 vs. 21.71	—
Choudhary et al.[9] [81]	2021	$n = 140$ (70 w/UF, 70 control)	Indian	Mean D3: 12.81 vs. 19.83	—
Al-azzawi et al.[9] [82]	2022	$n =$ Not reported (70 w/UF, control not reported)	Iraqi	—	—
Harmon et al.[9] [83]	2022	$n = 1610$ (control not reported)	Black American	—	—
Kumari et al.[9] [84]	2022	$n = 310$ (102 w/UF, 208 control)	Indian	Mean D3: 14.5 vs. 26.6	OR[7] 18,26 OR[7] 1.49
Chen et al.[9] [85]	2023	$n = 110$ (55 w/UF, 55 control)	Chinese	Mean D3: 20.5 vs 24.1	OR[8] 0.9
Rabia et al.[9] [86]	2023	$n = 101$ (16 w/UF, 85 control)	Pakistani	Mean D3: 26.3 vs. 32.6	—
Latifi et al.[9] [87]	2024	$n = 105$ (55 w/UF, 50 control)	Iranian	Mean D3: 16.02 vs. 15.99	—
Okoro et al.[9] [88]	2024	$n = 150$ (75 w/UF, 75 control)	Nigerian	Mean D3: 15.26 vs. 22.45	—

(*continued on next page*)

TABLE 2 (*continued*)

Author	Year	Patient cohort	Nationality	Mean D3 (UF vs. control), ng/mL	Odds ratio (OR)
Ahmad et al.[9] [89]	2024	$n = 160$ (80 w/UF, 80 control)	Indian	Mean D3: 10.85 vs. 19.64	—

Abbreviations: 25(OH)D3, 25-hydroxyvitamin D3; aOR, adjusted odds ratio; D3, vitamin D3; LDH, Lactate dehydrogenase; UF, uterine fibroid; w/UF, with uterine fibroids.

[1] aOR of fibroids with serum 25(OH)D3 < 10 ng/mL, adjusted for age, BMI, Black ethnicity, and parity.
[2] OR of fibroids with serum 25(OH)D3 > 20 ng/mL, adjusted for age and ethnicity.
[3] aOR of fibroids with serum 25(OH)D3 < 20 ng/mL, adjusted for race, age, parity, BMI, season, and physical activity.
[4] Crude OR of fibroids with 25(OH)D3 < 10 ng/mL.
[5] Crude OR of fibroids with 25 (OH)D3 < 30 ng/mL.
[6] aOR of fibroids with a higher concentration of 25(OH)D3, adjusted for BMI, gravity, parity, and LDH.
[7] Crude OR of fibroids with 25 (OH)D3 < 10 ng/mL. Crude OR of fibroids with 10 ng/mL < 25(OH)D3 < 20 ng/mL.
[8] aOR of fibroids adjusted for age, BMI, medical history of diabetes, and hypertension.
[9] Newly added studies.

concentrations in Black American women aged 23 to 35 y over the course of 5 y and found a 22% reduction in incidence in those with a serum vitamin D concentration ≥ 30 ng/mL; however, this result was not significant [adjusted odds ratio = 0.78; 95% CI: 0.47, 1.30]

Two of the 14 new studies added reported no significant differences in the vitamin D concentrations of women with fibroids when compared with control groups [76,87]. Ingala et al. [76] recruited 432 participants from different hospitals throughout the Democratic Republic of Congo and found that vitamin D deficiency was more frequently seen in patients with fibroids (17.1%), with significantly higher proportions when compared with controls (10.2%). However, the mean concentrations of serum vitamin D were similar in both groups (9.00 compared with 9.10 ng/mL). Still, vitamin D deficiency, along with other factors like age, family history of fibroids, and high serum progesterone concentrations, was found to be associated with increased risk of fibroids [76]. Latifi et al. [87] examined the concentrations of vitamin D, VDBP, and free vitamin D in 55 patients with fibroids and 50 healthy controls in Iran. The study found no significant differences in the levels of total vitamin D between the groups, but they found significantly lower levels of free vitamin D and higher VDBP in patients when compared with the control group [87].

B2. Clinical observational studies evaluating hypovitaminosis D in relation to fibroid size, quantity, and location

The relationship between vitamin D concentrations and fibroid size, quantity, and location remains unclear because findings have been inconsistent across studies. Out of 12 studies reporting serum vitamin D concentration and fibroid volume, 7 found an inverse relationship, including Sabry et al. [61], Srivastava et al. [67], Sarbhai et al. [71], Al-azzawi et al. [82], Kumari et al. [84], Chen et al. [85], and Okoro et al [88]. On the other hand, Paffoni et al. [62], Oskovi et al. [65], Singh et al. [66], and Masoem et al. [77] found no correlation. In addition, another study by Latifi et al. [87] found an inverse relationship between free vitamin D concentrations and fibroid size. Eight studies evaluated the relationship between fibroid number and serum vitamin D concentration, and 2 of these studies, Al-azzawi et al. [82] and Ahmad et al. [89], found an inverse correlation, whereas the remaining 6 found no correlation [62,65,66,69,84,85]. Finally, 2 studies examined the relationship between serum vitamin D concentrations and location of fibroids; however, none found a correlation [84,89].

C1. Meta-analysis: serum 25(OH)D concentrations in patients with fibroids compared with control

Mean 25(OH)D serum concentrations measured in ng/mL were reported in 3636 patients, 1752 of whom had been diagnosed with uterine fibroids, and the remaining 1884 of whom included subjects with a fibroid-free uterus, thus serving as the control group. Meta-analysis revealed significantly lower serum 25(OH)D concentrations in patients with fibroids compared with the control group (MD = –6.61 ng/mL; 95% CI: –8.83, –4.39). Heterogeneity tests revealed a Q value of 839.0959 ($P < 0.001$) and I^2 and τ^2 values of 97.5% and 23.3506, respectively (Figure 2A). Asymmetry was observed in the funnel plot of the meta-analysis, which was confirmed by Egger's test ($P = 0.0083$) but not in Begg's test ($P = 0.1209$), suggesting a potential publication bias (Figure 3A1). Trim-and-fill method using R_0 estimator found no missing on the right with an effect size of −6.6116 (95% CI: –8.7042, –4.5190) (Figure 3A2) [90]. Through sensitivity analyses (leave-one-out method and influence analysis), the study by Shanmugapriya et al. [70] (study #14) was identified as an outlier with Cook's D of 0.4497, and its exclusion resulted in a slight reduction of heterogeneity ($\tau^2_{del} = 13.1186$, Q_{del}: 762.5640) (Figure 4). The meta-analysis was repeated, excluding the outlier, and resulted in the following: (MD = –5.90 ng/mL; 95% CI: –7.63, –4.16) (Figure 2B). Another funnel plot was created after the removal of the outlier, and the trim-and-fill method using the R_0 estimator estimated 2 studies missing on the left with an effect size of −6.4604 ng/mL (95% CI: 8.1350, −4.7859) (Figure 3B1 and B2).

C2. Subgroup analysis of meta-analysis

A subgroup analysis based on the continent where the study took place revealed that most studies ($n = 14$) were conducted in Asia (MD = −6.46 ng/mL; 95% CI: −8.95, −3.97). There were 5 studies from Africa (MD = −5.30 ng/mL, 95% CI: −8.51, −2.08) and only 2 included from Europe (MD = −3.42 hg/mL, 95% CI: −15.18, 8.34) (Figure 5). Although all subgroups showed that patients with fibroids have lower concentrations of serum 25 (OH)D, this analysis shows that studies from Asia demonstrated a larger difference between the 2 groups.

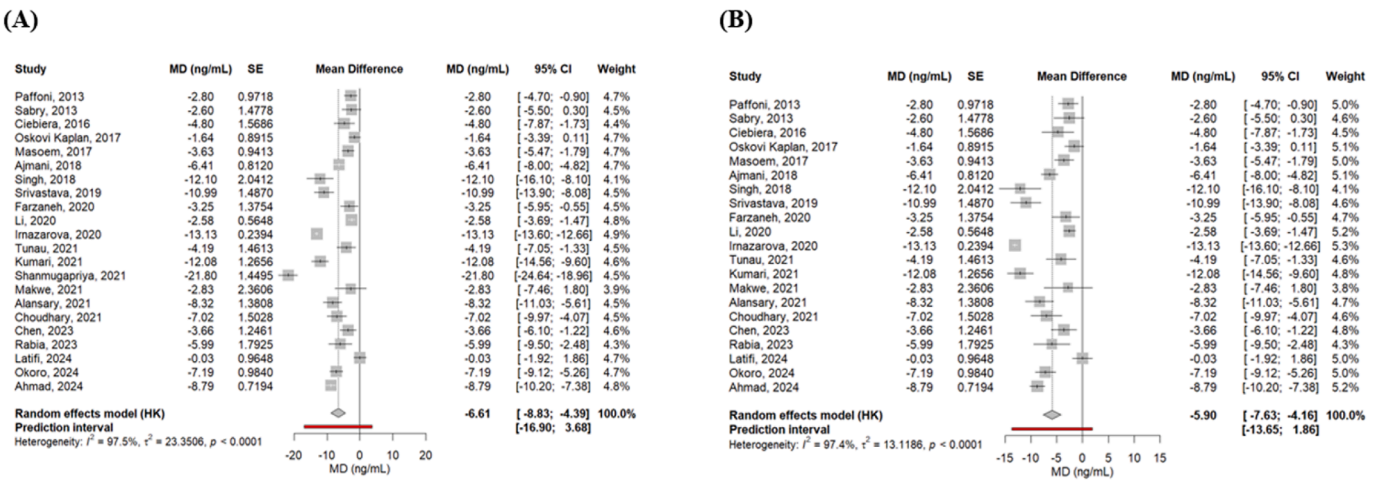


FIGURE 2. Forest plots of serum 25(OH)D concentrations MD between patients with uterine fibroids and controls; (A) before excluding outlier study, (B) after excluding outlier study. 25(OH)D, 25-hydroxyvitamin D; CI, confidence interval; HK, Hartung-Knapp adjustment; MD, mean difference.

C3. Metaregression of meta-analysis

The metaregression by publication date that was conducted suggests that the study findings tend to slightly decrease over time; however, the *P* value indicates that this downtrend is not statistically significant (slope = −0.2532, $P = 0.3459$) (Supplemental Figure 1). Thus, publication year does not affect the between-study variance ($R^2 = 0.00$). Supplemental Figure 1 shows that most of the studies are centered around the trend line.

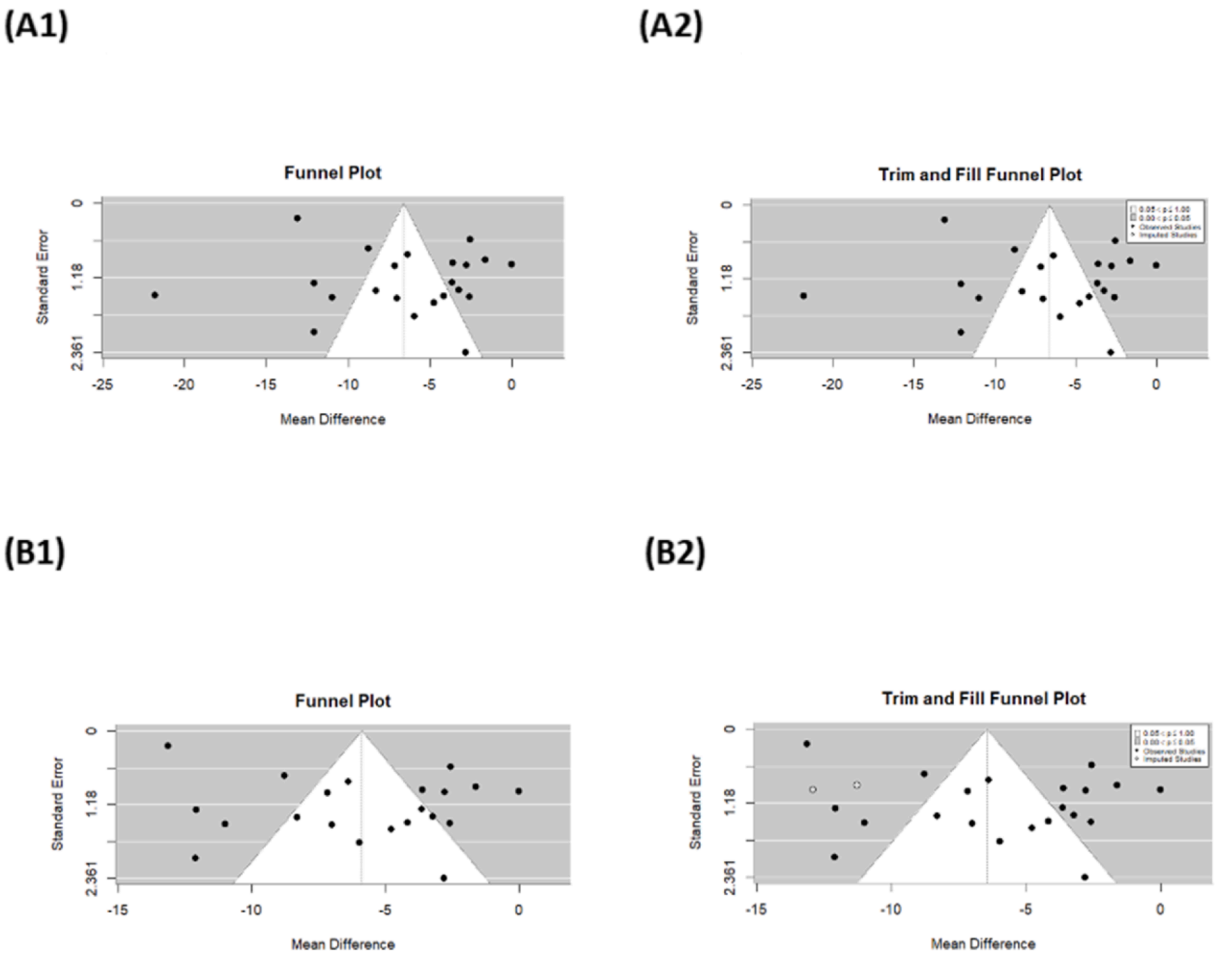


FIGURE 3. Funnel plots of serum 25(OH)D mean difference between patients with uterine fibroids and controls; (A) before excluding outlier study without R_0 estimator, (B) before excluding outlier study with R_0 estimator, (C) after excluding outlier study without R_0 estimator, (D) after excluding outlier study with R_0 estimator. 25(OH)D, 25-hydroxyvitamin D.

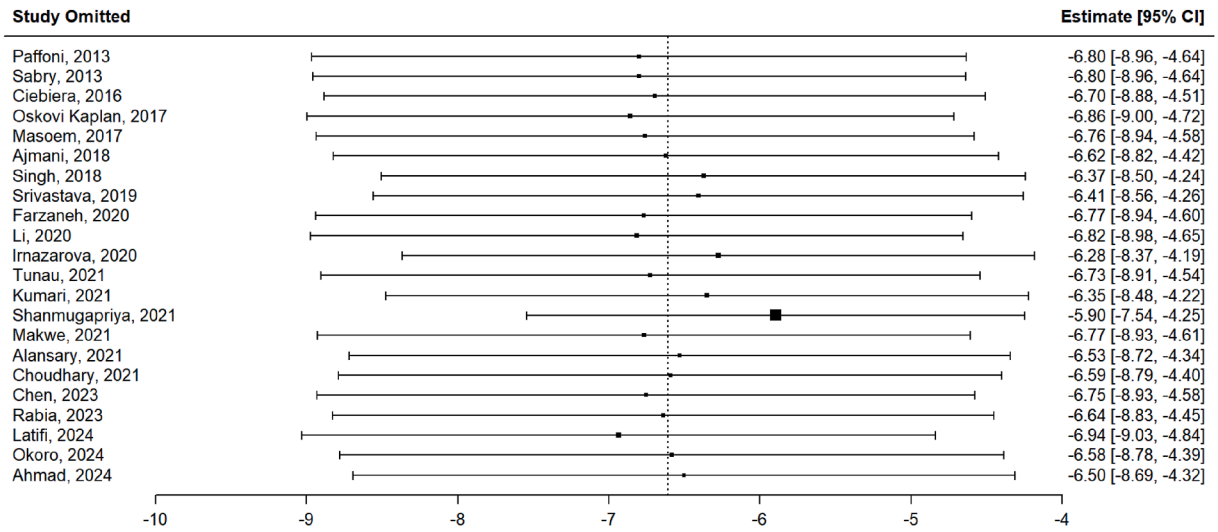


FIGURE 4. Sensitivity analysis for serum 25(OH)D concentrations MD between patients with uterine fibroids and controls (leave-one-out method). 25(OH)D, 25-hydroxyvitamin D; CI, confidence interval.

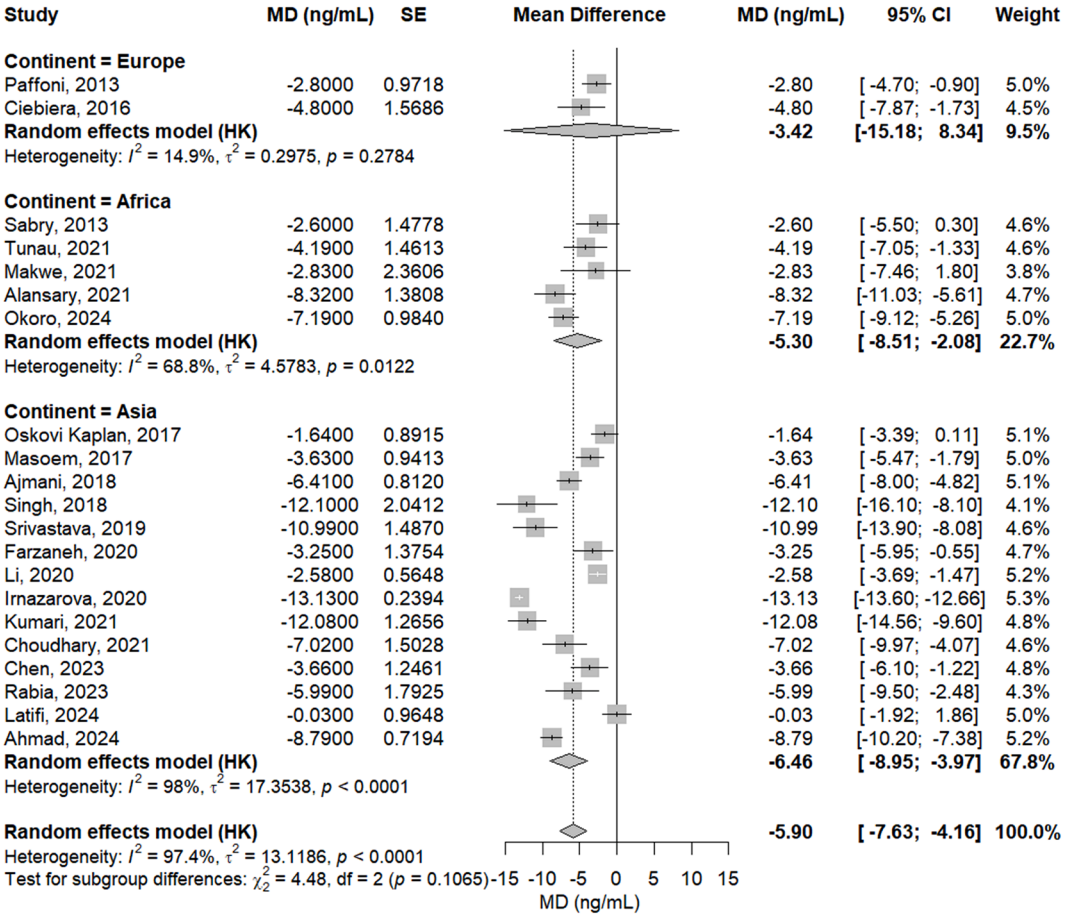


FIGURE 5. Subgroup analysis by continent for serum 25(OH)D concentrations MD between patients with uterine fibroids and controls (excluding outlier). 25(OH)D, 25-hydroxyvitamin D; CI, confidence interval; HK, Hartung-Knapp adjustment; MD, mean difference.

D1. Clinical interventional studies evaluating vitamin D for the treatment of uterine fibroids

Table 3 [26–28,83,91–101] shows a list of the clinical studies identified, with the main findings and characteristics for each. Five studies were initially identified through the past literature search that evaluated the use of vitamin D for the treatment of fibroids [26–29,91,92]. The new literature search yielded 4 more interventional studies [93–96] and 1 prospective cohort study [83] that evaluated vitamin D as a fibroid treatment. With the addition of the new studies identified through this review, a

TABLE 3
List of clinical studies that evaluate the use of vitamin D for the treatment of uterine fibroids

Author	Year	Inclusion	Exclusion	Intervention	Population	Results	Study type
Ciavattini et al. [91]	2016	Caucasian women, with ≥1 fibroids ≥ 10 mm diameter; serum vitamin D < 30 ng/mL	>4 fibroids, > 50-mm diameter fibroid, or severe fibroid-related symptoms	50,000 IU cholecalciferol weekly × 8 wk then 2000 IU daily for 1 y	108 women (53 in "study group," 55 control)	Significant increase in median fibroid diameter and volume in the control group, and no significant difference in the "study group"	Prospective cohort
Hajhashemi et al. [28]	2019	35—49 yo women, fibroid on USG > 2 cm in diameter; vitamin D < 20 ng/mL	Chronic medication use, or medical comorbidities, severe symptoms	50,000 IU every 2 wk × 10 wk	69 women (34 in treatment group, 35 in control)	Significant decrease in fibroid diameter in the treatment group	Randomized clinical trial
Arjeh et al. [26]	2020	18–45 y women, UF on USG with one fibroid > 10 mm in diameter; serum vitamin D < 30 ng/mL	>4 fibroids; hormonal therapy within 3 mo; vitamin D supplementation within 6 mo	50,000 IU vitamin D weekly × 12 wk	60 women, 22–40 y old (30 in treatment, 30 control)	Significant reduction in fibroid volume in the treatment group, with no significant reduction in control	Randomized clinical trial
Suneja et al. [92]	2021	Women with symptomatic fibroids with volume of > 2 cm^3 and vitamin D < 30 ng/mL	Pregnancy, lactation, adnexal pathology, and malignancy	60,000 IU vitamin D weekly × 8 wk then every other week × 8 wk	30 women, no control	No significant change in volume or number of fibroids over 8 or 16 wk	Prospective cohort
Tanha et al. [27]	2021	Premenopausal women with fibroid > 10 mm in diameter and vitamin D 21–29 ng/mL	Severe symptoms, hormonal treatment or vitamin D supplement within 3 mo, BMI < 18 or > 30 kg/m^2, malignancy	50,000 IU vitamin D weekly for 8 wk	204 women (102 in treatment, 102 controls)	Significant difference between treatment and control UF diameter change after 8 wk (−2.5% vs. 14.5%)	Randomized clinical trial
Tabrizian et al.[1] [93]	2021	35–45 y women with a history of single myoma and vitamin D ≤ 30 ng/mL	History of metabolic disease, multiple myomas, using hormonal or antihormonal drugs. History of cancer or premature ovarian failure	50,000 IU vitamin D weekly to vitamin D deficiency (<20 ng/mL) or vitamin D 1000 IU daily to vitamin D insufficiency (20–30 ng/mL) for 12 wk No control	137 women (85 with vitamin D deficiency, 52 with vitamin D insufficiency)	Significant increase in submucosal myomas. No significant decrease in myoma size in both the vitamin D insufficient and deficiency groups	Prospective cohort
Harmon et al.[1] [83]	2022	23–34 y women, self-identify as African American, intact uterus	Prior diagnosis of fibroids, cancer that requires radiation or chemotherapy, or lupus, Graves' disease, Sjogren's, scleroderma, or multiple sclerosis that requires medication	No intervention	1610 women	Significantly less growth rate associated with 25 (OH)D ≥ 20 ng/mL compared with 25(OH)D < 20 ng/mL. No significant difference in incidence or loss rate	Prospective cohort
Dawood and Sharief[1] [94]	2023	20–45 y women, presence of 1–3 fibroids	Pregnancy, breastfeeding, or delivery within 6 mo. History of myomectomy, loss of	50,000 IU vitamin D weekly for 10 wk. Followed by 2000 IU daily for 9–12 mo	66 women (43 women in treatment, 23 control)	Significantly increased vitamin D concentration and no difference in volume and diameter of fibroid for the	Prospective cohort

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TABLE 3 (*continued*)

Author	Year	Inclusion	Exclusion	Intervention	Population	Results	Study type
			pregnancy before 24 wk, hormone therapy, abnormal uterine, and difficult re-evaluation by ultrasound			experimental group. Significant increase in volume and diameter of fibroid for the control group	
Ayesha et al.[1] [95]	2023	30–50 y women with menorrhagia, premenopausal, and myomas 3 cm to 13 cm	Age < 30, Pregnancy, malignancy, myomas < 3 cm or > 13 cm	50,000 IU vitamin D, 500 mg Tranexamic acid, and 10 mg dydrogesterone weekly for 10 mo	111 women (no control)	Significantly increased vitamin D concentration, a decrease in the size of the uterine fibroid	Prospective cohort
Wang et al.[1] [96]	2024	30–48 y premenopausal women with uterine fibroid diameter between 1 and 6 cm. serum vitamin D < 30 ng/mL	Serum vitamin D > 30 ng/mL, receiving surgical treatment, suspected, or diagnosed hysteromyoma and adenomyosis, hormone within 3 mo, pregnancy, lactation, menopause, tumor, vitamin D supplementation within 1 y, history of autoimmune diseases	1600 IU vitamin D daily for 3 mo	64 women (32 in treatment, 32 controls)	Significant increase in serum vitamin D concentration, significant reduction in diameter, elastic strain ratio, and no significant change in blood flow	Randomized control trial
Studies with EGCG and vitamin D							
Porcaro et al. [97]	2020	≥18 y, premenopausal ≥ 2 cm^3 myoma with moderate myoma-related symptoms	Pregnant, breastfeeding, severe anemia, eligible for surgery, elevated liver function tests, estrogen, progesterone, corticosteroids, or supplements within 3 mo; sPRMs or GnRH analogs within 6 mo	EGCG 150 mg, vitamin B6 5 mg, and 25 μg vitamin D twice daily for 4 mo	30 women, 28–46 years old (15 in treatment, 15 control)	Significant reduction of myoma volume (34%) in the treatment group compared with an increase (7%) in controls; improved QOL in the treatment group	Randomized clinical trial
Miriello et al. [98]	2021	≥18 y, premenopausal, fibroid on USG < 4 cm in diameter	Pregnant, severe anemia, BMI >30, >5 fibroids, history of myomectomy, treatment with corticosteroids or hormones, elevated liver enzymes	EGCG 300 mg, vitamin B6 10 mg, and vitamin D 50 μg/d for 4 mo	95 women (41 treatment, 54 control)	Total volume and peripheral vascularization significantly decreased in the treatment group; a significant increase in volume and vascularization in the control	Randomized clinical trial
Grandi et al. [99]	2022	>40 y with uterine fibroid ≥ 3 cm diameter	Concomitant polyps, adenomyosis, or suspected malignancy; use of hormonal treatments	EGCG 300 mg, vitamin B6 10 mg, and vitamin D 50 μg/d for 90 d	16 women, 43–52 y old, no control group	Significant reduction in fibroid diameter	Prospective cohort

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TABLE 3 (*continued*)

Author	Year	Inclusion	Exclusion	Intervention	Population	Results	Study type
Tinelli et al.[1] [100]	2024	30–40 y with uterine fibroid > 4 cm, eligible for laparoscopic myomectomy due to symptoms	pregnancy, breastfeeding, BMI >30, diabetes, suspected or diagnosed malignancy, use of supplements, use of hormonal treatments	300-mg EGCG, 50 μg vitamin D, 10 mg vitamin B6, 50 mg inositol daily for 3 mo	91 women 44 treatment, 47 control	Significant reduction in the duration of surgery and blood loss during surgery	Randomized clinical trial
Vitamin D for the prevention of fibroids							
Vahdat et al. [101]	2022	30–45 y, BMI 18.5–24.9, uterine submucosal fibroids, vitamin D concentration 20–30 ng/mL	Underlying medical condition, other gynecologic condition, use of hormonal treatment within 3 mo, use of vitamin D supplement within 6 mo; remaining fibroids after myomectomy	Hysteroscopic myomectomy followed by 1000 IU vitamin D daily for 12 mo	109 (55 treatment, 54 control)	Lower recurrence rate in treatment group (9.1% compared with 18.5%, $P = 0.17$); recurrent fibroids were significantly smaller in the treatment group (13.5 vs. 21.2 mm diameter)	Randomized clinical trial

Abbreviations: EGCG, epigallocatechin gallate; GnRH, gonadotropin-releasing hormone; QOL, quality of life; sPRMs, selective progesterone receptor modulators; UF, uterine fibroid; USG, ultrasound.

[1] Newly added studies.

total of 10 studies involving 2460 patients were included. Of the newly added clinical studies, Ayesha et al. [95] and Wang et al. [96] showed that raising vitamin D concentrations to normal parameters significantly reduced fibroid size, and Harmon et al. [83] showed a lower fibroid growth rate associated with higher vitamin D concentrations. Meanwhile, Dawood and Sharief [94] observed that supplementation could stabilize but not reduce fibroid size. In addition, only Tabrizian et al. [93] found no association between adequate vitamin D concentrations and the treatment of fibroids.

In the study by Ayesha et al. [95], 50,000 IU of vitamin D together with 500 mg of tranexamic acid and 10 mg of dydrogesterone was prescribed orally every week for 10 mo to 111 Pakistani premenopausal women with menorrhagia and fibroids (3 cm–13 cm in size). The study showed a significant increase in vitamin D concentrations ($\bar{d} = +17.90$ ng/mL; SD: 8.94) and a significant decrease in fibroid size ($\bar{d} = -4.84$ cm; SD: 1.55) after the intervention, supporting the association between vitamin D and a reduction in fibroid size [95]. A similar result was seen in the study by Wang et al. [96], which recruited 64 Chinese women with fibroids (1–6 cm in diameter) and with serum vitamin D concentrations < 30 ng/mL. The experimental group ($n = 32$) received 1600 IU of vitamin D per day, and the control group ($n = 32$) received a placebo. After 3 mo, the experimental group showed a significant increase in vitamin D concentrations ($\bar{d} = +6.69$ ng/mL) and a significant reduction in the mean diameter ($\bar{d} = -0.66$ cm) and elastic strain ratio ($\bar{d} = -0.37$) of the fibroids. In addition, clinical symptoms improved overall across the experimental group after the intervention, with a significantly decreased incidence of urinary frequency ($\bar{d} = -25\%$), dysmenorrhea ($\bar{d} = -21.9\%$), and heavy menstrual bleeding ($\bar{d} = -21.9\%$). None of these symptoms improved in the control group [96]. The findings of these studies support those observed in Hajhashem et al. [28] and Arjeh et al. [26], 2 of the clinical studies from the past review. In addition, in the prospective community cohort study by Harmon et al. [83], the serum vitamin D concentrations in 1610 self-identified Black American women without a previous diagnosis of fibroids were measured over the span of 5 y. The study reported that those with vitamin D concentrations ≥ 20 ng/mL were associated with a reduced fibroid growth rate and had 9.7% less growth per 18 mo when compared with those with concentrations < 20 ng/mL. In addition, those with concentrations ≥ 30 ng/mL had 32% higher fibroid loss between-study visits than those with concentrations < 30 ng/mL (RR: 1.32; 95% CI: 0.95, 1.83) [83].

Dawood and Sharief [94] also evaluated the effect of vitamin D supplementation on fibroid size in 66 Iraqi women with vitamin D deficiency, with the experimental group ($n = 43$) receiving 50,000 IU weekly for 10 wk followed by 2000 IU daily for 9 to 12 mo, whereas the control group ($n = 23$) received a placebo. The results showed significantly increased vitamin D concentrations ($\bar{d} = +17.1$ ng/mL) but no difference in the size of fibroids in the experimental group. However, there was a significant increase in fibroid diameter ($\bar{d} = 5$ mm) and volume ($\bar{d} = 2.9$ cm^3) in the control group. In addition, no differences in fibroid number, site, or location were found in either group [94]. These results resemble those from Ciavattini et al. [91], Suneja et al. [92], and Tanha et al. [27], which were identified

in the past review, and suggest that vitamin D supplementation may be useful in halting fibroid growth but may not be as effective in reducing the size of existing fibroids.

Finally, the study by Tabrizian et al. [93] included 137 Iranian women with a history of a single fibroid and a vitamin D concentration $\leq$ 30 ng/mL. These women received 50,000 IU weekly for 12 wk if identified as vitamin D deficient ($<$ 20 ng/mL) and 1000 IU weekly for 12 wk if insufficient (30–20 ng/mL). Notably, the women within the vitamin D–deficient group had a significantly higher initial fibroid size when compared with those in the insufficiency group. However, the results found that a significant increase in submucosal fibroids was observed, as well as no significant decrease in fibroid size in either group after the intervention. These findings do suggest that supplementation may not be effective for short-term treatment of fibroids; however, the study may be limited by the short interventional exposure time [93].

D2. Clinical interventional studies evaluating the treatment of fibroids with vitamin D in combination with epigallocatechin gallate

Epigallocatechin gallate (EGCG) is the main component found in green tea and has gained interest for its role in promoting health due to its antineoplastic, antioxidant, and antifibrotic effects [102,103]. Three previously identified clinical studies evaluated vitamin D treatment in conjunction with EGCG [97–99]. One more study by Tinelli et al. [100] regarding EGCG was also identified and included for a total of 232 participants. The previously identified studies all took place in Italy, used the same supplement composition (EGCG 300 mg, vitamin B6 10 mg, and vitamin D 50 μg/d), and all showed a significant decrease in fibroid size after the intervention [97–99]. The study by Tinelli et al. [100] also took place in Italy and used a similar treatment plan, but also added 50 mg of D-chiro-inositol. The inclusion criteria included women who would be undergoing a laparoscopic myomectomy because of the severity of their fibroid-related symptoms, and treatment efficacy was evaluated based on surgical outcomes. Those who received the intervention showed a significantly reduced surgery duration and reduced blood loss during surgery [100].

D3. Clinical interventional study on vitamin D for the prevention of fibroids

The previous paper reported that Vahdat et al. [101] evaluated vitamin D for secondary prevention of fibroids and showed a lower recurrence risk of fibroids (9.1% compared with 18.5%, $P = 0.17$) and a smaller diameter in recurrent fibroids (13.5 compared with 21.2 mm, $P < 0.001$) for women who received vitamin D supplementation after the removal of fibroids through myomectomy when compared with those who did not receive supplements after surgery [29]. No new studies were identified that looked at the use of vitamin D for the prevention of fibroids.

Discussion

The main findings of the study include that patients with uterine fibroids have significantly lower concentrations of serum vitamin D when compared with women with a “normal,” fibroid-free uterine structure, and that, overall, vitamin D has been associated with an improvement of fibroid-associated symptoms and potentially aids in decreasing fibroid size or halting fibroid progression. This was revealed both by the meta-analysis and by the systematic review of studies involving the association of vitamin D with uterine fibroids.

In 2013, 3 studies were among the first to identify this association, and the findings of 23 studies published between 2015 and 2024 further confirmed this relationship [61–63]. Apart from these, 3 different studies reported no significant differences in vitamin D concentrations between women with fibroids and controls; however, Ingala et al. [76] still found vitamin D deficiency to be a risk factor for fibroids, and Latifi et al. [87] revealed significantly lower concentrations of free vitamin D in patients with fibroids [75]. Although the study by Mitro and Zota [75] found no correlation, its results may have been influenced by the study’s reliance on self-reported data instead of the use of ultrasounds to ascertain the presence of fibroids. In contrast, 22 (95.7%) out of the 23 studies that found an association reported the use of ultrasonography for assessing fibroid status, making these findings more reliable. In addition, the mean serum vitamin D concentrations varied widely among these studies in women with fibroids (ranging from 6.54 ng/mL in Oskovi et al. [65] to 34.2 ng/mL in Makwe et al. [79]), as well as in women without uterine fibroids (ranging from 8.18 ng/mL in Oskovi et al. [65] to 37.1 ng/mL in Makwe et al. [79]). This variation highlights the impact of factors that influence vitamin D concentrations, such as geographical location, sunlight exposure, diet, and skin pigmentation, on study outcomes.

Vitamin D is obtained from sunlight exposure through a process where it is synthesized in the skin from 7-dehydrocholesterol (7-DHC), a derivative of cholesterol. UV radiation converts 7-DHC into previtamin D3, which isomerizes into vitamin D3 through a nonenzymatic, temperature-dependent process and is ultimately released into the circulation [17,19,104]. As a biologically inert molecule, vitamin D3 requires further processing to become active. In the liver, it is converted into 25(OH) D3, which is used as a biomarker for circulating vitamin D concentrations. It is then further metabolized in the kidney to form the biologically active molecule, 1,25-dihydroxyvitamin D3 [37,105]. The observed effect of vitamin D on fibroids may be explained by multiple different mechanisms through which vitamin D is able to attenuate fibroid growth; these were explored by various studies through in vitro models. Markowska et al. [51] found that VDR expression was diminished in uterine muscle tissue at the periphery of fibroids in comparison with healthy uterine muscle and that there were significantly fewer immunopositive cells in both the fibroids and the peripheries when compared with normal uterine tissue, indicating a lower expression of VDRs. This further supports the association between vitamin D and fibroid growth and suggests that this downregulation of receptors may be a mechanism for fibroid development. Other studies found that vitamin D is able to interact with different proteins, allowing it to mediate processes such as cell proliferation, fibrosis, apoptosis, and DNA repair [16,24,36–50]. Therefore, a decrease in vitamin D concentrations has been found to result in a lack of regulation of these processes, thus contributing to the formation and growth of uterine fibroids. In addition, genetic polymorphisms regarding vitamin D and VDRs were also explored. Different studies found certain genetic variations, such as the DHCR7 rs1044482 C $>$ T

polymorphism [57] and the TT genotype of the *Fok1* polymorphism [54], to be associated with an increased risk for fibroids.

From the random-effect model, women with fibroids had a pooled effect of −6.61 ng/mL, indicating lower vitamin D concentrations when compared with those in the control group. The 95% prediction interval of −8.83 ng/mL to −4.39 ng/mL signifies that the true mean difference of vitamin D serum concentration is negative and substantial. However, results from Cochran's Q test for heterogeneity (Q value = 839.0959; $P < 0.001$) suggested that there is significant variation in true effect size among studies, and I^2 and τ^2 values show that other clinical or methodological factors between studies are contributing to this heterogeneity, other than just sampling error. After the identification and removal of the outlier study, which contributed to large heterogeneity, the meta-analysis was rerun to gain a more precise estimate of the effect size. Despite removing the outlier, a large negative pooled difference of −5.90 ng/mL effect, with a 95% CI of −7.63 to −4.16, still suggested the effect size was significantly negative. It should be noted that there was still large heterogeneity among in-between-study variability ($I^2 = 97.4\%$; $P < 0.001$) after removing considerable total heterogeneity ($\tau^2_{del} = 13.1186$, Q_{del}: 762.5640).

The geographical distribution of the studies was assessed through a subgroup analysis by continent, and some variations in the pooled effect sizes were observed across subgroups. Of the 22 studies, 14 (66.7%) were conducted in Asia, 5 studies (23.8%) were from Africa, and only 2 studies (9.52%) came from Europe. Studies from Asia showed a large difference between the study and control groups with a pooled MD of −6.46 ng/mL, whereas studies from Africa (MD = −5.3 ng/mL) and Europe (MD = 3.42 ng/mL) only showed a moderate difference. This variation between subgroups was likely due to the small number of studies from different continents, which resulted in less representative results. Notably, <10% of the studies were conducted in Europe, which may be attributed to the disproportionate impact of uterine fibroids on people of color, which ultimately influences the geographical distribution of research efforts. A more internationally representative population may have provided more consistent data. Nevertheless, all findings were statistically significant, and the results consistently indicated lower concentrations of vitamin D in patients with fibroids across all subgroups. In addition, the metaregression did not show any significant association between the effect size and publication year of the papers studied.

An important limitation of the meta-analysis was the high heterogeneity between the included studies that could not be addressed, even after removing the outlier study. Although the geographical subgroup analysis explained some of the variability among studies, there are limitations to only modeling at the continental level. Most studies had a diagnosis of uterine fibroids confirmed with sonography, which may adjust for one factor; however, there were often significant differences in underlying population age, BMI, and study vitamin D assay types across studies, all of which may have their own impact on heterogeneity. Moreover, factors such as sunlight exposure, parity, and the season of blood sampling were often unreported. Because of the lack of stratified data, these factors could not have been taken into consideration. As established in a recent expert consensus, individualized vitamin D supplementation strategies are essential given that vitamin D status is affected by a variety of demographic, clinical, and biological factors, resulting in differing dosing requirements across populations. Therefore, to accurately assess the effects of supplementation, future studies should incorporate stratification based on relevant patient characteristics [106]. Overall, the substantial heterogeneity observed across studies, which may reflect differences in study design, population characteristics, and unmeasured or unreported confounders, limits the generalizability and interpretation of the pooled difference in serum vitamin D concentrations. A more internationally representative population that explores vitamin D serum concentration with stratification of other risk factors may help provide more consistent data that may allow for a better understanding of vitamin D's role in uterine fibroids. Another important limitation of the meta-analysis was the potential small study bias and publication bias present, as seen from the Egger's test. Although the trim-and-fill funnel plot (after outlier removal) showed that there may be an estimated 2 potentially missing studies from the left side, examining the adjusted effect size (MD: −6.4604 ng/mL) and improved CI (95% CI: −1.05, −0.31) from the trim-and-fill plot and literature review suggested that the lower serum vitamin D uterine fibroid compared with control was still significant, and vitamin D supplementation showed a promising role in regulating uterine fibroids.

Current clinical evidence has supported the use of vitamin D in improving uterine fibroid–related symptoms; however, because of differing results in the studies, it is still unclear whether the use of vitamin D supplementation aids in reducing the size of existing fibroids or in simply preventing further growth and formation. Nevertheless, 9 of the 10 (90%) clinical studies evaluating the use of vitamin D for the treatment of fibroids found vitamin D to be efficient in either reducing or stabilizing fibroid size [26–28,83,91,92,94–96]. In addition, 1 study reported significant improvements in fibroid-associated symptoms after 3 mo of vitamin D supplementation [96]. Recommendation guidelines should consider the current evidence available and encourage individuals diagnosed with or at risk of fibroids to have their vitamin D concentrations regularly screened to receive the necessary supplementation if found deficient. Moreover, clinical studies evaluating the use of vitamin D in combination with EGCG as treatment also revealed promising results in reducing fibroid size and in yielding better surgical outcomes during myomectomies [97–100]. Further studies, including RCTs, are needed to confirm whether vitamin D has true therapeutic efficacy in the treatment or prevention of fibroids.

In conclusion, a meta-analysis of 21 studies demonstrated a large negative effect size, indicating that women with fibroids had markedly lower vitamin D concentrations compared with healthy controls. Although substantial variability was found between the studies, the results remained significant. This association was further confirmed throughout the systematic review, with findings consistently supporting this link. In addition, evidence from clinical studies has shown that vitamin D supplementation can be associated with improvement of fibroid-associated symptoms and may play a role in reducing fibroid size or halting further progression of existing fibroids. Future studies should seek more internationally representative study populations and should focus on the role of vitamin D in fibroid prevention.

Acknowledgments

The writing team would like to acknowledge the formatting work done by Tala Qasem regarding this manuscript.

Author contributions

The authors' responsibilities were as follows – BS, JS: devised the project, helped supervise the project, and primarily responsible for the final content; AIG-C, BS: drafted the manuscript; HK, LP, MA, OY, MSI, AA-H, JS: reviewed the manuscript critically; HK, LP: helped with the data extraction; and all authors: contributed to the data analysis of the final manuscript, approved the final version to be published, and took responsibility for the integrity of the data and the accuracy of the data analysis.

Conflict of interest

The authors report no conflicts of interest.

Funding

This study was supported by a grant titled "Reproductive Medicine Collaborative Consortium: a randomized placebo-controlled trial of epigallocatechin gallate to improve fertility in women with uterine fibroids" from the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD), R01 HD100365

Data availability

The data that support the findings of this study are available in the supplementary material of this article. Any additional data is available from the corresponding author upon reasonable request.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

The authors declare that no generative AI or AI-assisted technologies were used in the writing of this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tjnut.2026.101838.

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